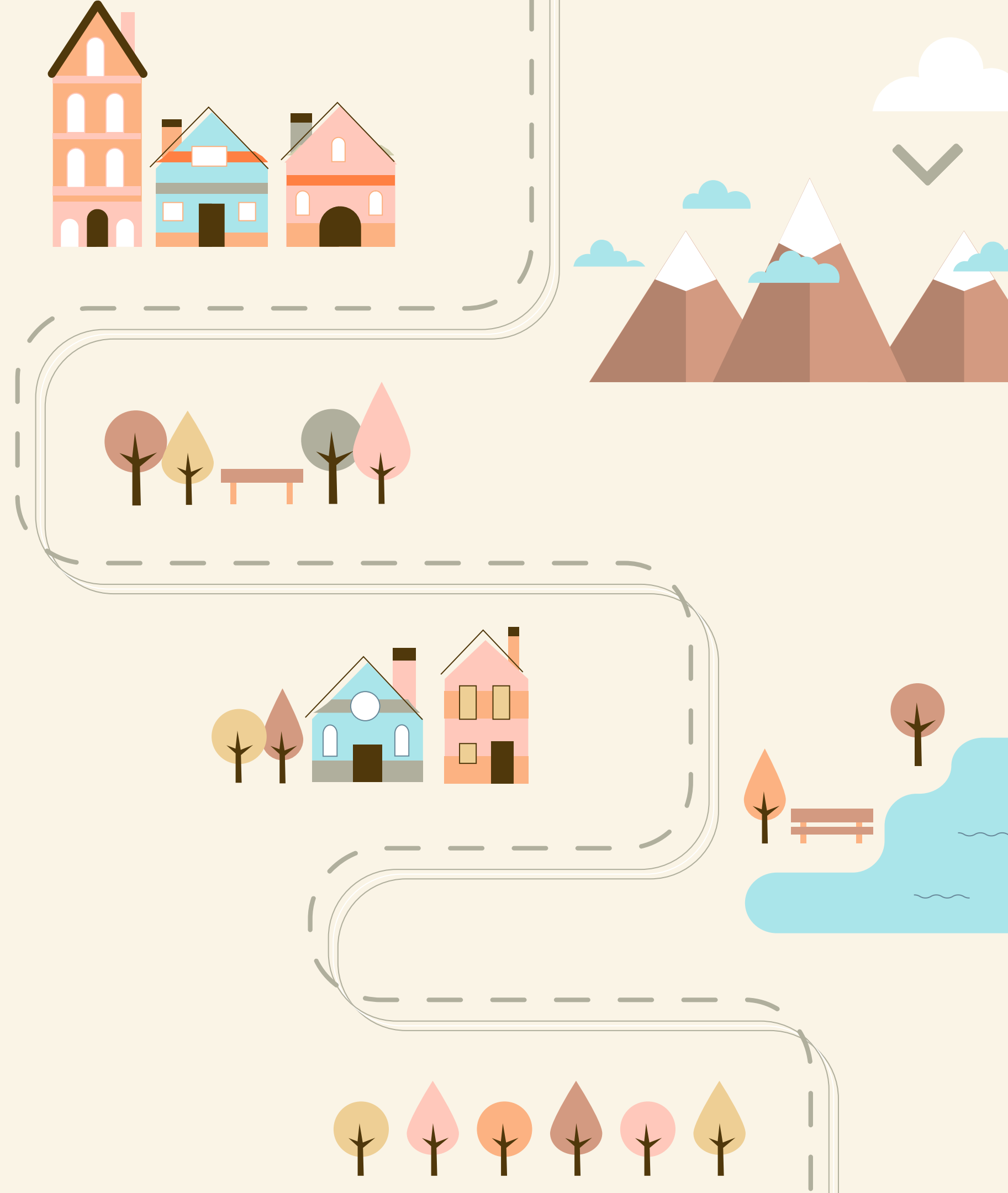


Cape Town AirBnb Price Prediction

Data-Driven Pricing for Cape Town Market

by Group 11





Meet Our Team

Kenneth Kiplagat

Project Lead

Warren Patsi

Technical Lead

Ivy Mbugua

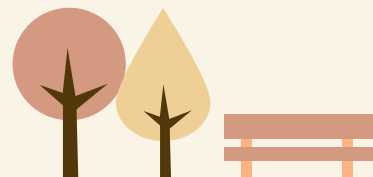
Data & Modeling Lead

Dennis Chesire

Deployment Engineer

Mercy Kusetei

Data Analyst



The Host's Pricing Dilemma

Hosts face 3 key challenges in Cape Town's competitive market:



Neighborhood Challenge

Location matters, standardized pricing fails across Cape Town's diverse neighborhoods



Property Challenge

Each property requires custom pricing based on its specific features and offerings.



Revenue vs. Demand Balance

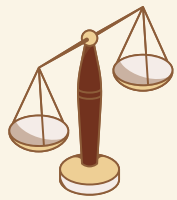
Set prices too high and you get fewer bookings. Set them too low and you lose money

What We Aimed To Solve

Primary Objective:

- Build a smart pricing system that tells Cape Town hosts exactly what to charge and why

OUR TARGETS:



Compare Models:

- Find the most accurate pricing algorithm for Cape Town's unique market



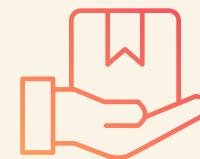
Engineer Features:

- Identify what really drives prices: location, amenities, seasonality



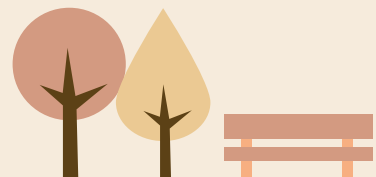
Ensure Explainability

- Explain exactly why each price is recommended



Deliver a Tool:

- Create a simple web app where hosts get instant pricing recommendations



How We Built The Model

Data Preparation

1

Cleaned missing data,
limited extreme prices,
and created 20+ useful features.

Performance Evaluation

3

Reached 80% accuracy,
Kept pricing errors around 23%,
Performed reliably across neighborhoods

5

Data Understanding

Sourced 20,000+ Cape Town listings,
Studied price and feature trends,
Extracted insights

Model Selection

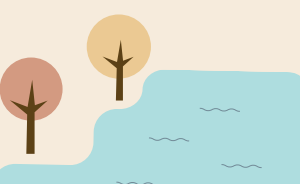
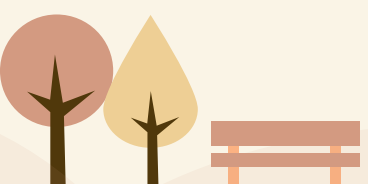
2

Explored six model options,
Tuned the top performers,
selected XGBoost

4

Deployment

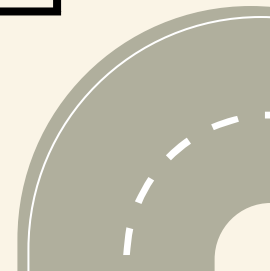
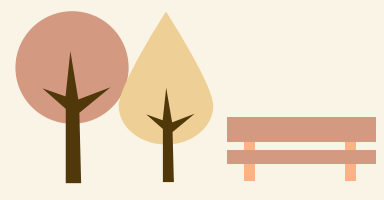
Built a real-time pricing app for hosts,
this improves mobile accessibility,
ready for expansion to other cities



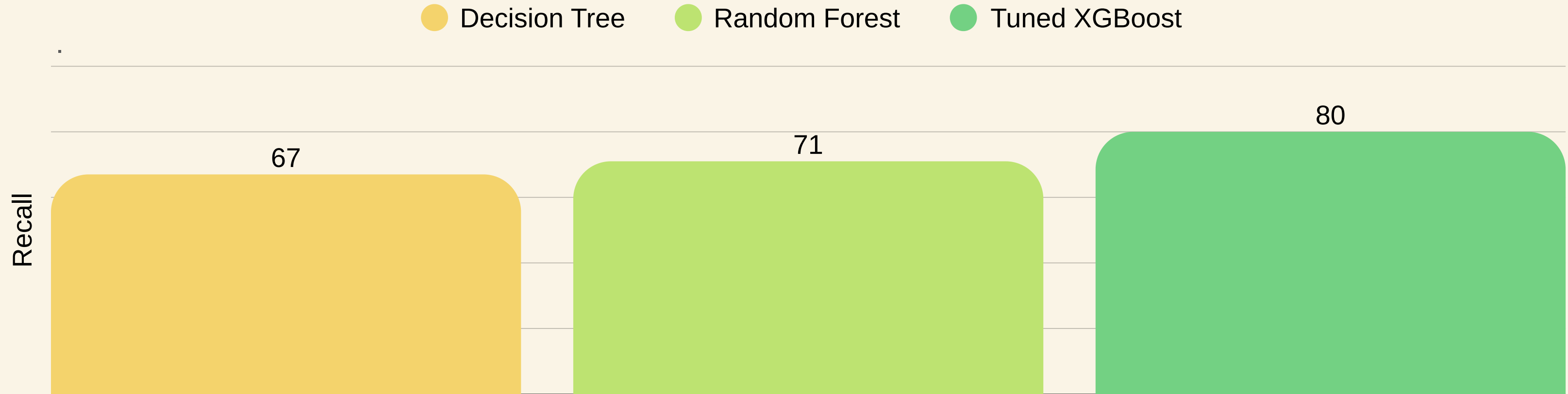
What Fed Our Model

<u>Data Category.</u>	<u>Variables</u>	<u>Description</u>
LOCATION	neighbourhood_cleansed	Where in Cape Town? your neighborhood sets your price range.
PROPERTY DETAILS	Type, Bedrooms, Accommodates	How big is it? size and capacity of your property determines the base price.
HOST QUALITY	Superhost Status, host experience, Response Rate	Are you a trusted host? Reliability justifies premium prices
AMENITIES	Pool, Wi-Fi, Ocean View, Parking	What special features? Luxury amenities allows you to charge more
TARGET VARIABLE	price	The ultimate question: What's the right price for your listing?

Source: *Inside Airbnb* | **Snapshot:** *28 Sept 2025* | **Core File:** *listings.csv.gz*



Finding Our Best Predictor



Decision Tree

- A good foundation that might need small adjustments
- Solid starting point for your listings



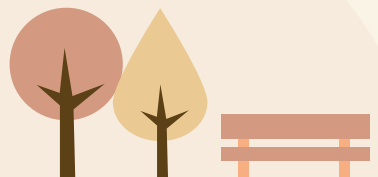
Random Forest

- Considers multiple factors for better accuracy
- Gives you confident, competitive rates

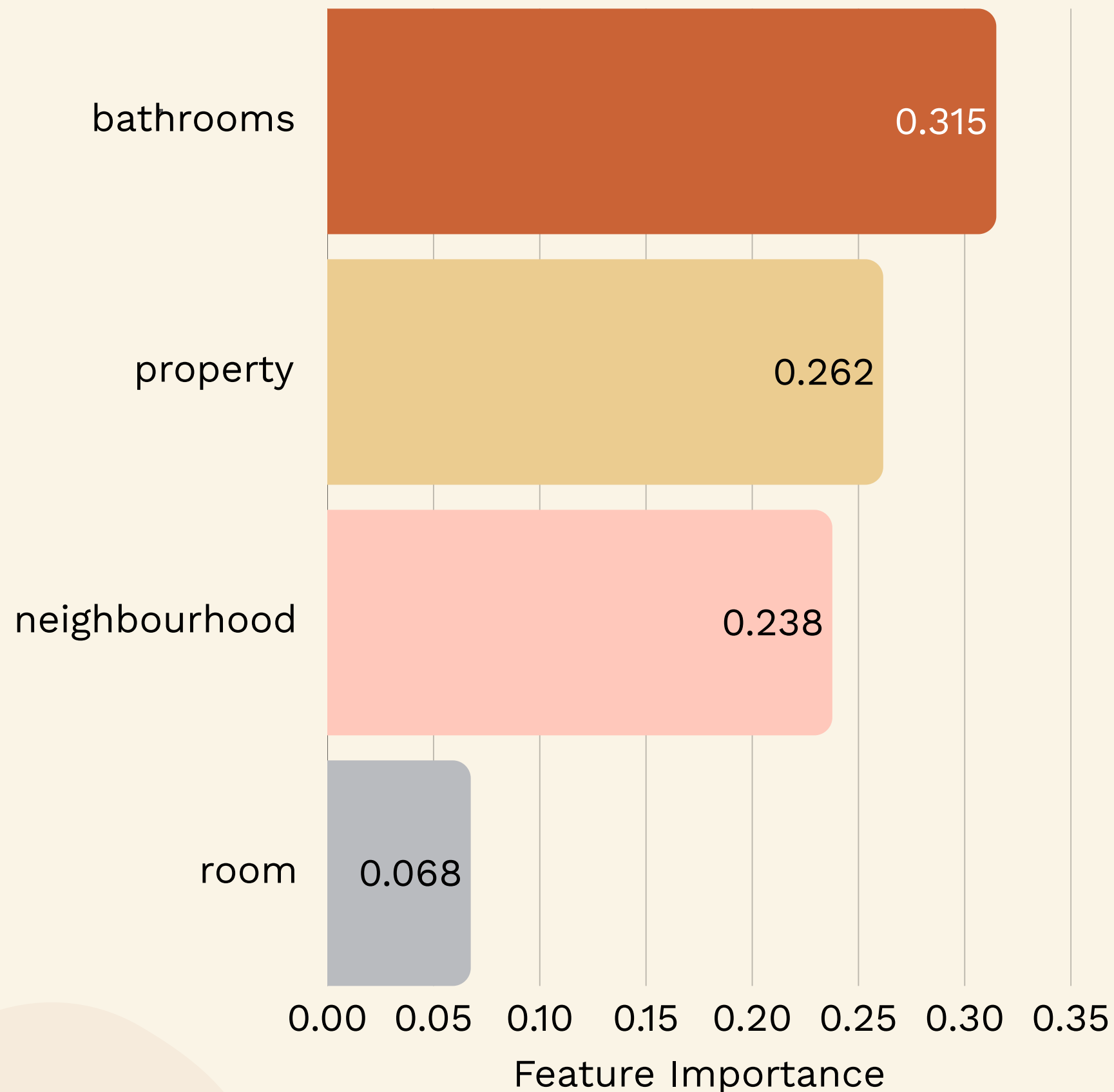


Tuned XGBoost

- Most precise pricing for maximum profit
- Deeply understands Cape Town's market nuances
- Delivers premium accuracy that puts more money in your pocket



What Truly Drives The Price



Maximize Bathroom Value

- Ensure listings highlight the number of bathrooms, as they significantly influence guest preferences and pricing.

Leverage Property Type for Pricing

- Tailor your pricing based on property type (e.g., apartment vs. house) to attract the right customers.

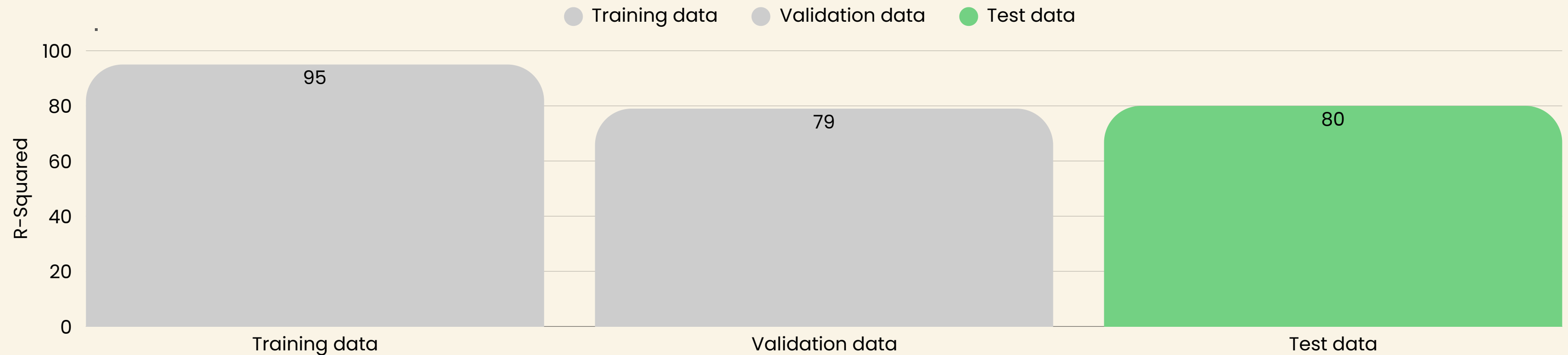
Capitalize on Neighborhood Demand

- Prioritize high-demand neighborhoods like City Bowl and Atlantic Seaboard to justify premium pricing.

Highlight Room Type for More Guests

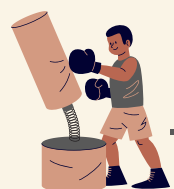
- Emphasize entire homes for higher pricing, as they attract larger groups and families.

How The Model Performed



Robustness

- The model performs relatively well on new, unseen properties as it does on the data it was trained on
- The minimal performance drop proves the model captures genuine market trends, not just memorized data

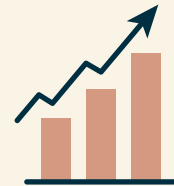


Consistency

- With nearly identical results on validation (79) and test (80) data, you get reliable pricing every time



What We Achieved



Strong Predictive Performance

- Our tuned XGBoost model achieved R^2 of 0.80, demonstrating high accuracy in price predictions.



Actionable Host Insights

- The model provides explainable recommendations, helping hosts understand pricing decisions.



Key Pricing Drivers Identified

- Location, property type, and capacity emerged as the most influential factors for optimal pricing.



Practical Solution Delivered

- We successfully transitioned from analysis to a functional web tool for real-world application.

See The Model In Action



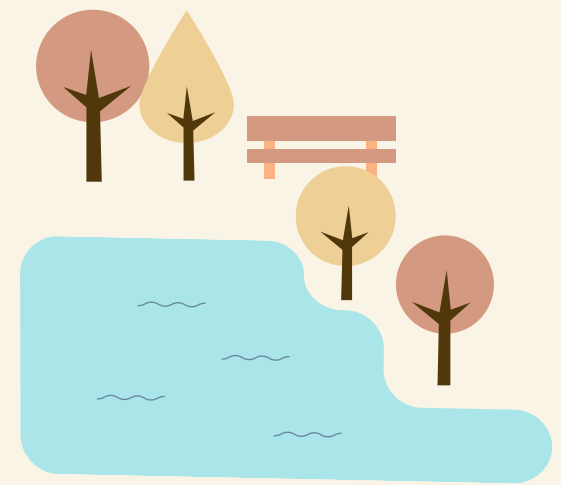
Deployed & Ready to Use

Real-time web application
Accessible on any device
Instant sentiment predictions

 **Link**

Features

Test with any listing features
See confidence scores for each prediction
Understand model decision-making



Where We Go From Here



Adaptive Model

Continuously retrain the model with new market data to maintain pricing accuracy over time.



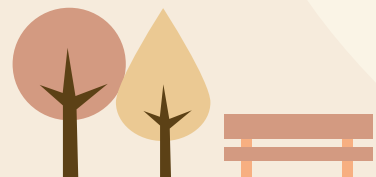
Host Education

Equip hosts with insights on key pricing drivers to enable informed and confident decision-making.



Geographic Expansion

Scale the successful model to new cities (nairobi) and gather host feedback for further refinement.



Thanks!

Do you have any questions?

Github

Price Prediction App

